

Geospatial Foundation Models Benchmarking

Datasets and Evaluation Challenges

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1 Current State of GFM Benchmarking Literature

Corley et al. [1] audited 152 papers related to Geospatial Foundation Models (GFMs) and argue that the current literature is substantially less comparable and reproducible than individual papers may suggest. Their study provides a systematic snapshot of the current GFM evaluation landscape.

The corpus also highlights some ambiguity in the use of the term “foundation model”. Only 46% of the 152 papers explicitly describe their proposed model as a foundation model in the title, abstract, or contributions. The remaining papers are earlier self-supervised remote-sensing models that previous GFM surveys already consider part of the same research lineage. The authors therefore treat the 46% figure as descriptive rather than as a criterion for inclusion.

The corpus was seeded from previous GFM surveys and expanded through two additional searches: an OpenAlex and Semantic Scholar citation-graph expansion, aimed mainly at recovering 2024–2025 papers not covered by the surveys, and a keyword search over remote-sensing self-supervised learning papers published between 2019 and 2023, before “foundation model” became common terminology in the field.

The resulting corpus contains 152 papers published between 2019 and 2025. Papers from 2026 were excluded because the year was incomplete at the time of submission. Metadata extraction was LLM-assisted and subsequently manually validated by the authors [1].

1.1 Model Weights Are Often Unavailable

Across the corpus, 39% of papers release no model weights [1]. A further 19% provide a public code repository but no released model artifact, meaning that reuse would require attempting to reproduce the original pretraining procedure.

This represents a basic obstacle to reproducible benchmarking. Before a model can be evaluated on a common downstream benchmark or used as a rerun baseline, the corresponding pretrained checkpoint must first be available.

1.2 The Field Does Not Converge on a Shared Core of Benchmarks

The 152 papers report evaluations on 401 distinct benchmarks, corresponding to 1,046 evaluation experiments in total [1].

The three most frequently used benchmarks—EuroSAT [2], NWPU-RESISC45 [3], and AID [4]—account for only 10.6% of all evaluations. The remaining 89.4% is distributed across 398 benchmarks, most of which appear in only one or two papers.

The problem is not simply that some benchmarks are more popular than others. Rather, individual papers frequently evaluate on almost disjoint benchmark sets.

For each paper, Corley et al. [1] compute the fraction of its downstream benchmarks that overlap with the ten most commonly used benchmarks in the corpus. The mean overlap is only 0.27, and 50 out of 143 papers, approximately 35%, have no overlap at all with the top ten.

At the corpus level, benchmark usage has a Gini coefficient of approximately 0.51. Moreover, this coefficient remains approximately stable after 2023, while the number of benchmarks appearing in only one paper increased from 13 in 2022 to 98 in 2025. The evidence therefore does not suggest convergence towards a shared evaluation core.

This creates a GFM-specific *benchmark lottery*: because models are rarely evaluated on sufficiently overlapping benchmark sets under common protocols, the published literature does not provide enough evidence to construct a reliable field-wide ranking of GFMs [1].

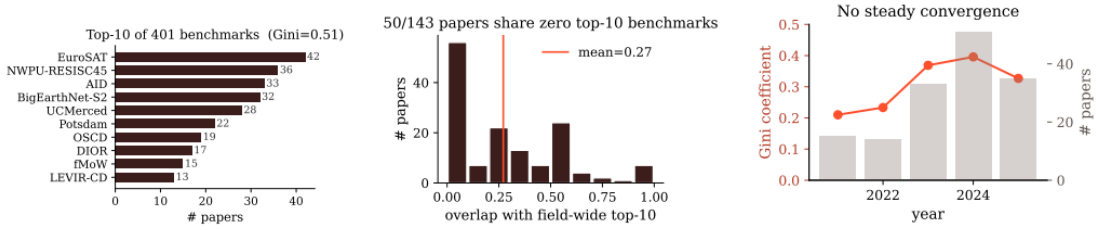


Figure 1: Benchmark usage across the audited GFM literature. The figure reports the most frequently used downstream benchmarks, per-paper overlap with the field’s top-10 benchmarks, and the evolution of benchmark concentration over time. Adapted from Corley et al. [1].

1.3 Reported Metrics Can Diverge Substantially Under the Same Nominal Protocol

The authors extracted every (model, benchmark, metric, evaluation protocol, training regime) tuple from the corpus, obtaining 10,817 results after benchmark and metric normalisation [1].

Evaluation protocols were grouped into categories such as fine-tuning, linear probing, kNN probing, zero-shot, and few-shot evaluation.

After filtering, 301 model–benchmark–protocol combinations appeared in at least two different papers. For each combination, the authors compute

$$\text{spread} = \max(\text{reported score}) - \min(\text{reported score}).$$

Of these 301 combinations, 76 have a spread of at least 5 percentage points, 46 at least 10 points, and 20 at least 20 points [1].

The largest discrepancy concerns Scale-MAE [5] evaluated on NWPU-RESISC45 using a linear probe. One paper reports an accuracy of 33.0, whereas the original Scale-MAE paper reports 89.6, corresponding to a difference of 56.6 percentage points, despite the use of the same released ViT-L checkpoint and the same nominal evaluation protocol [1].

This example illustrates that the label “linear probe” does not uniquely specify an experiment. Hidden choices may include the optimiser, learning rate, number of epochs, image resizing or cropping, normalisation, data augmentation, representation layer, train/validation/test split, class subset, sensor channels, and other adaptation details.

A second example reported by the authors concerns GPT-4o on UCMerced under zero-shot evaluation, where reported accuracy ranges from 43.5 to 88.8 percentage points. The corresponding papers do not describe the prompting configuration in sufficient detail to explain the discrepancy [1].

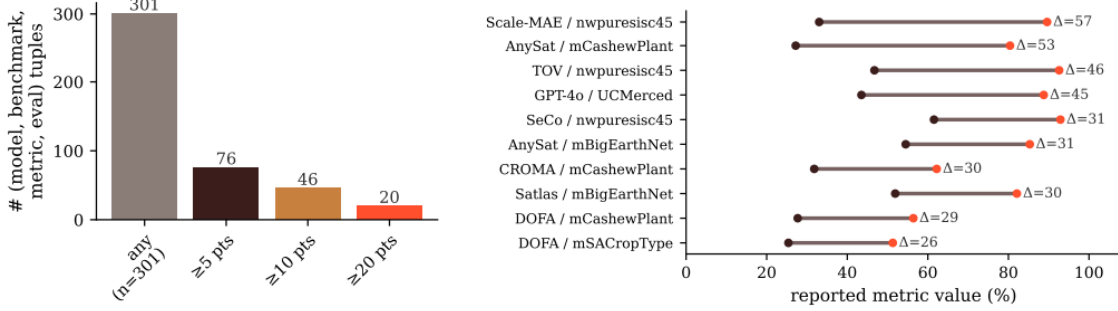


Figure 2: Largest cross-paper disagreements for repeated model–benchmark–protocol combinations. Even when the model, benchmark, and nominal evaluation protocol are the same, reported results can differ substantially. Adapted from Corley et al. [1].

Could This Be Explained by Random-Seed Variation?

The authors argue that random-seed variation alone is unlikely to explain the largest discrepancies.

Most repeated experiments agree closely, with a median spread close to zero, but the 90th-percentile spread is 12.7 percentage points [1].

Such a large tail is substantially greater than the run-to-run variation normally expected from simple classification heads under otherwise fixed conditions. The authors therefore interpret the largest disagreements primarily as evidence of hidden protocol differences.

1.4 Lack of a Shared Evaluation Harness

The geospatial community already provides benchmark collections and software infrastructure such as GEO-Bench [6], PANGAEA [7], and TorchGeo [8].

However, Corley et al. [1] argue that these resources do not yet constitute a community-wide evaluation harness comparable to those used in more mature evaluation ecosystems.

A benchmark collection may provide datasets, predefined splits, dataloaders, and example training scripts while still leaving substantial choices to the researcher. Consequently, two papers may both state that they evaluated on the same benchmark while actually executing different downstream experiments.

A stricter evaluation harness would additionally specify versioned task definitions, reference evaluation protocols, metrics, and automated checks, allowing a result to be reproduced from a model identifier and a task configuration.

Conceptually,

model → shared evaluation harness → standardised result.

Such a harness can reduce protocol-related degrees of freedom, but it cannot guarantee that the benchmark itself measures the intended deployment capability.

In particular, spatial dependence between training and test samples can inflate remote-sensing accuracy estimates [9, 10]. A perfectly reproducible evaluation may therefore still provide a misleading estimate of geographic generalisation.

Standardisation should also not necessarily imply forcing numerically identical hyperparameters on every architecture. Different models may require different optimisation settings. A fair comparison may instead require a common validation procedure and a comparable hyperparameter-search or compute budget, while allowing the final selected hyperparameters to differ between models.

1.5 Architecture and Pretraining Data Are Confounded

Another source of ambiguity arises when a new model simultaneously changes its architecture, pretraining data, and learning algorithm.

If several factors change at once, the observed downstream improvement cannot be attributed to one of them in isolation [1].

This is not an argument for requiring all GFMs to use identical pretraining data. Different, larger, private, or more diverse datasets may themselves be important contributions.

The problem arises when a paper attributes an improvement to an architectural or algorithmic modification without providing a controlled experiment in which the pretraining data are held fixed.

Corley et al. [11] show that apparent advantages of pretrained remote-sensing models over supervised ImageNet baselines can shrink substantially once preprocessing and distributional factors are controlled. The *No One Knows* study uses this as evidence that architecture and pretraining-data effects can be difficult to disentangle [1].

Similarly, Kaur et al. [12] study the effect of pretraining-data composition and show that changing the pretraining dataset alone can shift downstream performance by margins comparable to those often attributed to architectural novelty.

The corpus-level statistics make this comparability problem particularly clear.

Among the 126 papers for which a specific pretraining dataset could be extracted, the authors identify 87 distinct named primary pretraining datasets. Only 32 of the 126 papers, approximately 25%, share their complete pretraining configuration with at least one other paper. The remaining 94 use a configuration not shared by any other work in the corpus [1].

Consequently, comparisons in the GFM literature often involve systems that differ simultaneously in architecture, pretraining data, training objective, and downstream evaluation protocol.

Such comparisons remain useful at the *system level*, but they cannot by themselves establish which individual design choice caused a performance improvement.

1.6 Recommendations

Corley et al. [1] propose six recommendations aimed at improving comparability across the field:

1. **Release reusable model weights**, or explicitly state why they cannot be released.
2. **Evaluate on a minimum shared core of benchmark tasks**, while adding task-specific evaluations when needed.
3. **Distinguish copied and independently rerun baseline results**, including the corresponding source or configuration.
4. **Report uncertainty or repeated-run variability** when computationally feasible.
5. **Develop a shared third-party evaluation harness** with versioned protocols, common task definitions, and automated checks.
6. **Disentangle pretraining data, architecture, and learning algorithm** through controlled comparisons when causal claims are made.

2 Benchmark Datasets

2.1 Dataset Availability

Several efforts have already been made to catalogue, organise, and provide access to Earth Observation datasets for machine learning applications.

Schmitt et al. [13] provide a broad review of datasets developed for deep learning in Earth Observation, discussing their evolution across different sensors, tasks, spatial resolutions, and application domains. The same work introduces the IEEE GRSS Earth Observation Database (EOD), a community-driven catalogue intended to centralise metadata about publicly available remote-sensing datasets.

The EOD [14] provides an interactive search interface through which datasets can be filtered according to properties such as sensor modality, task, geographic location, dataset name, and keywords. This makes it a useful starting point for identifying candidate datasets satisfying specific experimental requirements.

A complementary effort is EarthNets [15], which reviews more than 500 publicly available remote-sensing datasets and organises them according to attributes including research domain, resolution, size, and relationships between datasets. Based on these attributes, the authors additionally select five large-scale datasets to form a benchmark and release the EarthNets platform for consistent model evaluation.

TorchGeo [8] provides a different but particularly useful resource from an experimental perspective. Rather than acting mainly as a dataset catalogue, TorchGeo implements a large collection of geospatial datasets through a common PyTorch interface, including benchmark datasets for classification, regression, semantic segmentation, object detection, instance segmentation, time-series analysis, and change detection. Many datasets can therefore be downloaded, loaded, and integrated into training pipelines with substantially less dataset-specific preprocessing code. This makes TorchGeo particularly relevant when selecting datasets that must later be evaluated across multiple models under a common experimental pipeline.

Additional community-maintained suites include `awesome-satellite-imagery-datasets` [16] and the `satellite-image-deep-learning/datasets` repository [17]. These resources are primarily useful for dataset discovery rather than as standardised benchmarking frameworks.

2.2 Candidate Benchmark Datasets

Starting from a broader inventory of Earth Observation datasets collected from the scientific literature, benchmark suites, dataset catalogues, and public repositories, datasets were progressively filtered according to their suitability for downstream foundation-model evaluation.

The initial screening distinguished datasets with documented use as downstream benchmarks from pretraining corpora, dataset collections, and resources not designed primarily for comparative evaluation. Confirmed benchmark datasets were then assessed according to practical and scientific criteria relevant to the planned experiments, including public availability, storage requirements, availability of downstream labels, geographic information and variability, pre-defined or constructible evaluation splits, compatibility with Earth Observation foundation models, and previous use in foundation-model evaluation.

This process produced the following set of 38 candidate datasets and benchmark resources for more detailed analysis. The list is not intended to constitute an exhaustive catalogue of all Earth Observation benchmarks. Rather, it represents the datasets identified through the sources considered in this study and retained after applying the above criteria. Other established or recently introduced datasets may therefore fall outside the inventory, and their absence should not be interpreted as an assessment of their scientific quality or relevance to Earth Observation research.

#	Dataset	Task	Modality	Storage (GB)	Geo	Splits	FM	Use	License
1	So2Sat LCZ42	C	S1+S2	51.8	Yes	Yes	High	High	CC-BY-4.0
2	CropHarvest	C	Multi	22.7	Likely	Likely	High	Medium	?
3	FMoW	C	RGB+MS	~3500	Yes	Yes	?	High	Yes
4	FMoW-WILDS	C	RGB	55	Yes	Yes	Partial	High	Yes
5	fMoW-Sentinel	C	S2 MS	~80	Partial	Yes	High	High	?
6	BigEarthNet v2.0 (S1+S2)	C,S	S1/S2	110	Yes	Yes	High	High	CDLA-Permissive-1.0
7	Open-Canopy	R,CD	SPOT-6/7+LiDAR	~360	Yes	Yes	Partial	High	?
8	DFC2020	S	S1/S2	10.6	Partial	Yes	High	High	?
9	Sen1Floods11	S	S1/S2	14.3	Yes	Yes	High	High	?
10	SpaceNet-7	CD,S	MS+MT	20.6	Yes	Yes	High	High	?
11	TimeSen2Crop	S	S2	1.1	Likely	Likely	High	High	?
12	PASTIS-R	S	Multiple (Satellite)	54	Yes	Yes	High	High	?
13	DynamicEarthNet	CD,S	PlanetFusion	524	Yes	Yes	Medium	High	?
14	HLS Burn Scars	S	HLS	2.65	Partial	Yes	High	High	?
15	Fields Of The World [paper]	S,I	S2 MSI	~4.0	Likely	Likely	High	High	?
16	Kuro Siwo	S	S1 SAR+DEM	7.53	Yes	Yes	High	High	CC-BY
17	MADOS	S	S2 MS	5.35	Yes	Yes	High	High	?
18	SEN12MS-CR	SR	S1/S2	389	Yes	Yes	High	High	?
19	SEN12MS	S,C	MS+SAR	510	?	?	High	High	?
20	South Africa Crop Type	C	S1+S2	82.8	Likely	Likely	High	Medium	?
21	Canadian Cropland	C	S2	26	?	?	High	High	?
22	TreeSatAI	C	Multi	16.3	?	?	High	High	?
23	m-Brick-Kiln (GEO-Bench)	C	S2 MS	1.6	?	Yes	High	High	?
24	m-ForestNet (GEO-Bench)	C	Landsat-8 MS	1.2	?	Yes	High	High	?
25	SICKLE	S,C	S1/S2+L8 TS	6	?	?	High	High	?
26	BioMassters	R	S1/S2	201	Yes	Yes	High	High	?
27	Chesapeake Land Cover	S	RGB+MS	140	?	?	High	Medium	?
28	South Sudan Crop Type	S	S1/S2+Planet	977	Yes	Yes	High	High	?
29	MARIDA	S	MS (Satellite)	1.2	?	?	High	High	?
30	CloudSEN12	S	S2/S1+DEM	~1000	Yes	Yes	High	High	?
31	Five-Billion-Pixels	S	RGB+MS	104	Yes	Yes	High	High	?
32	AI4SmallFarms	S	S2 MS	1.4	Yes	Yes	High	High	?
33	BraDD-S1TS	CD,S	S1 TS	17.7	?	?	High	High	?
34	DW-Expert [paper]	S	S1/S2	~10	Partial	Partial	High	High	?
35	CropSeg	S	HLS MS+MT	11.8	Yes	Yes	High	High	CC-BY-4.0
36	xView2	CD	RGB	51	Yes	Yes	Partial	High	?
37	WorldStrat	SR	Multi	110	Yes	?	High	Medium	Mixed
38	WILDS (benchmark suite)	B	Multi	-	Mixed	Yes	?	Reference	Mixed

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